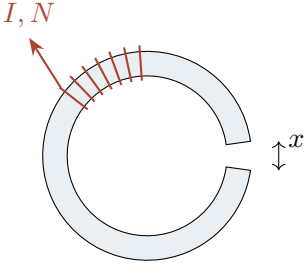
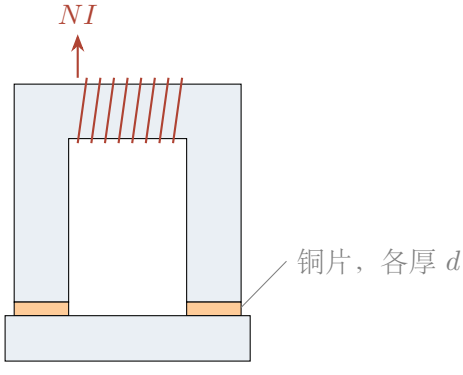
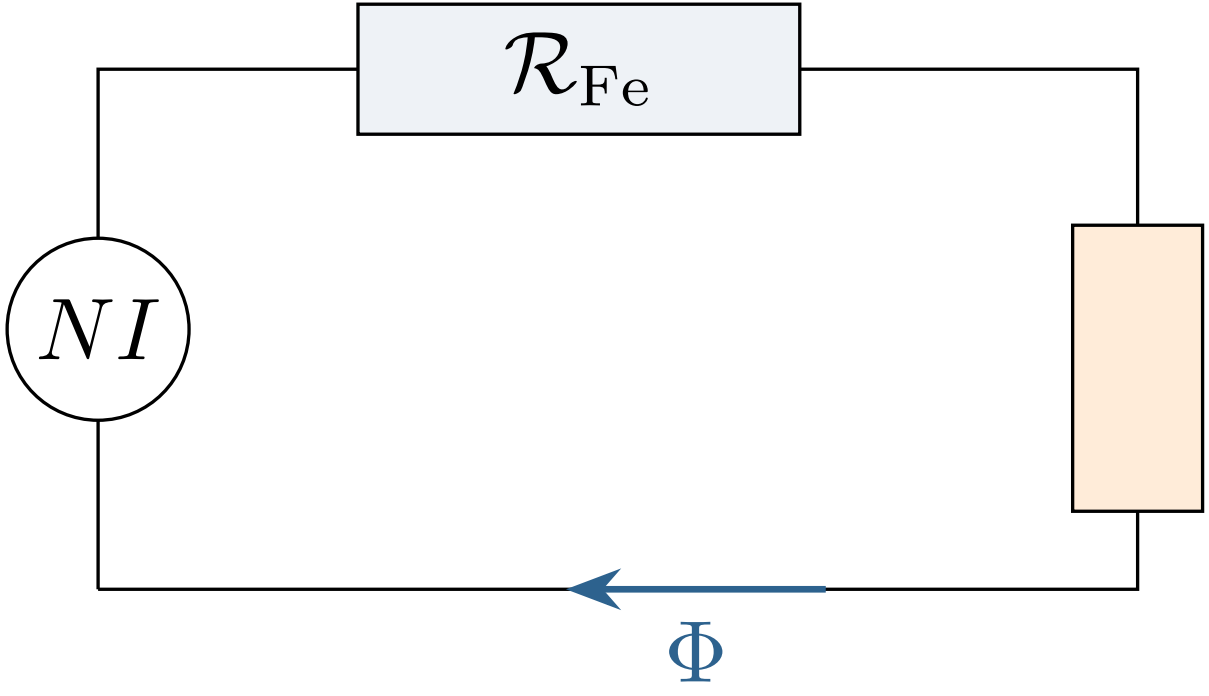


铁环：铁材长度 L，截面 A



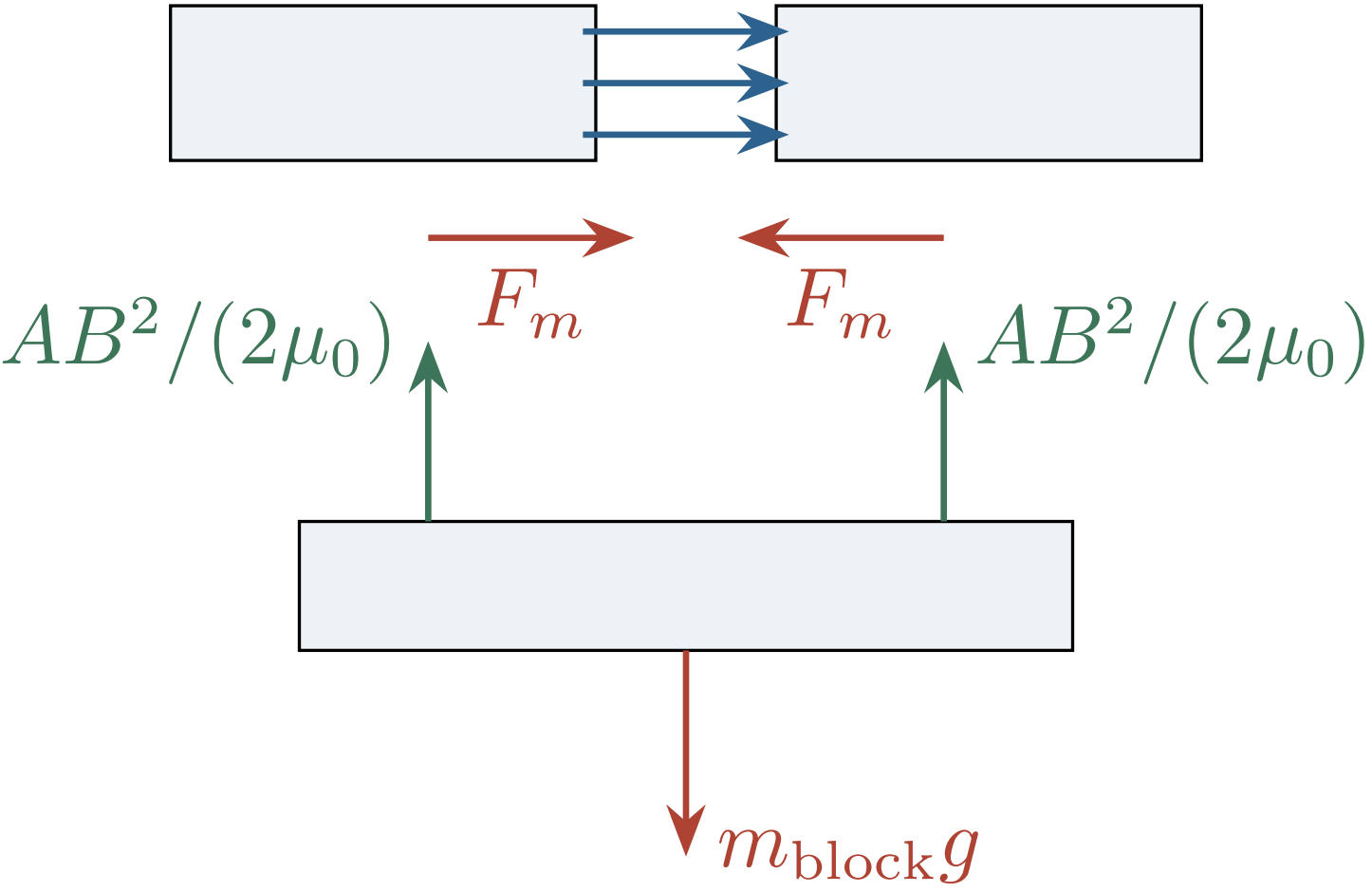
铁块



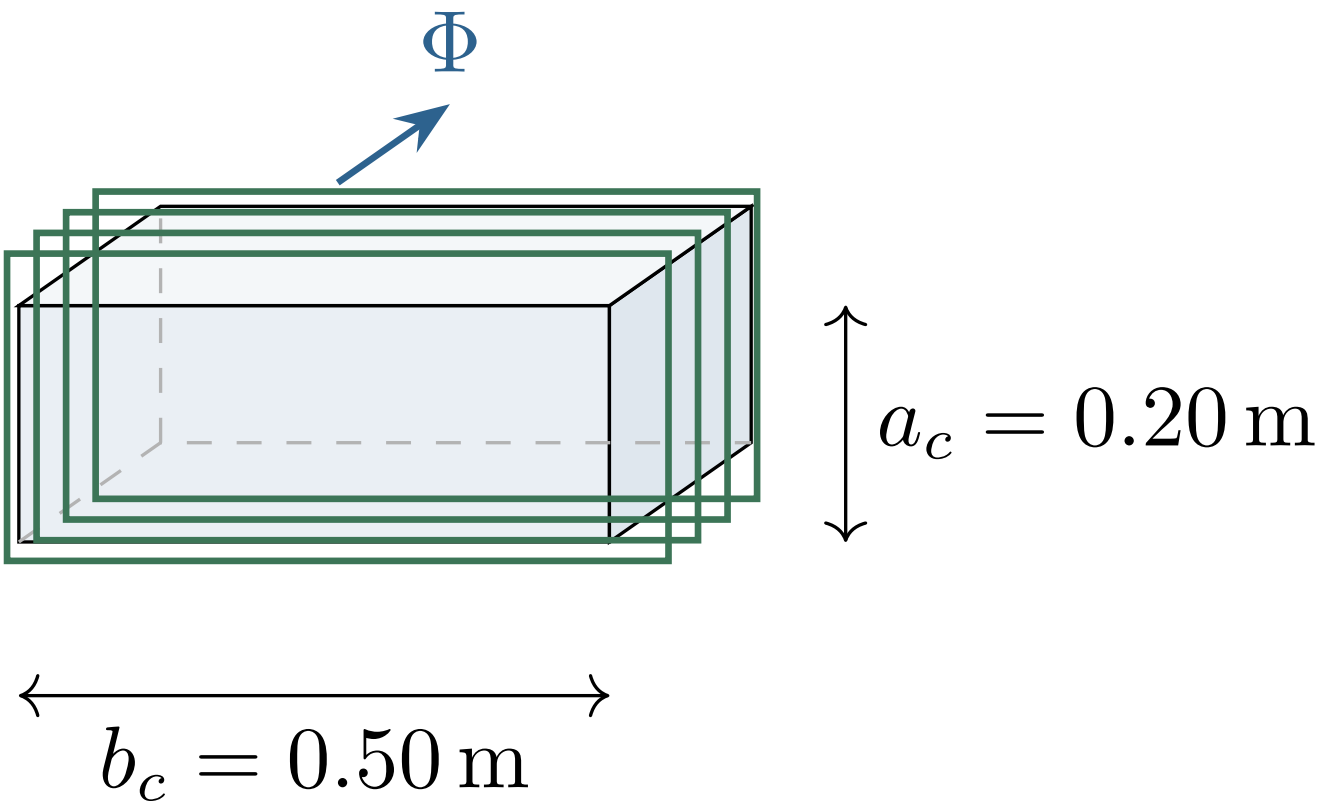
单间隙：$\mathcal{R}_{\mathrm{tot}} = \mathcal{R}_{\mathrm{Fe}} + \mathcal{R}_g$

双铜片：$\mathcal{R}_{\mathrm{tot}} = \mathcal{R}_{\mathrm{Fe}} + 2d/(\mu_0 A)$

同一间隙，两端的相互作用力

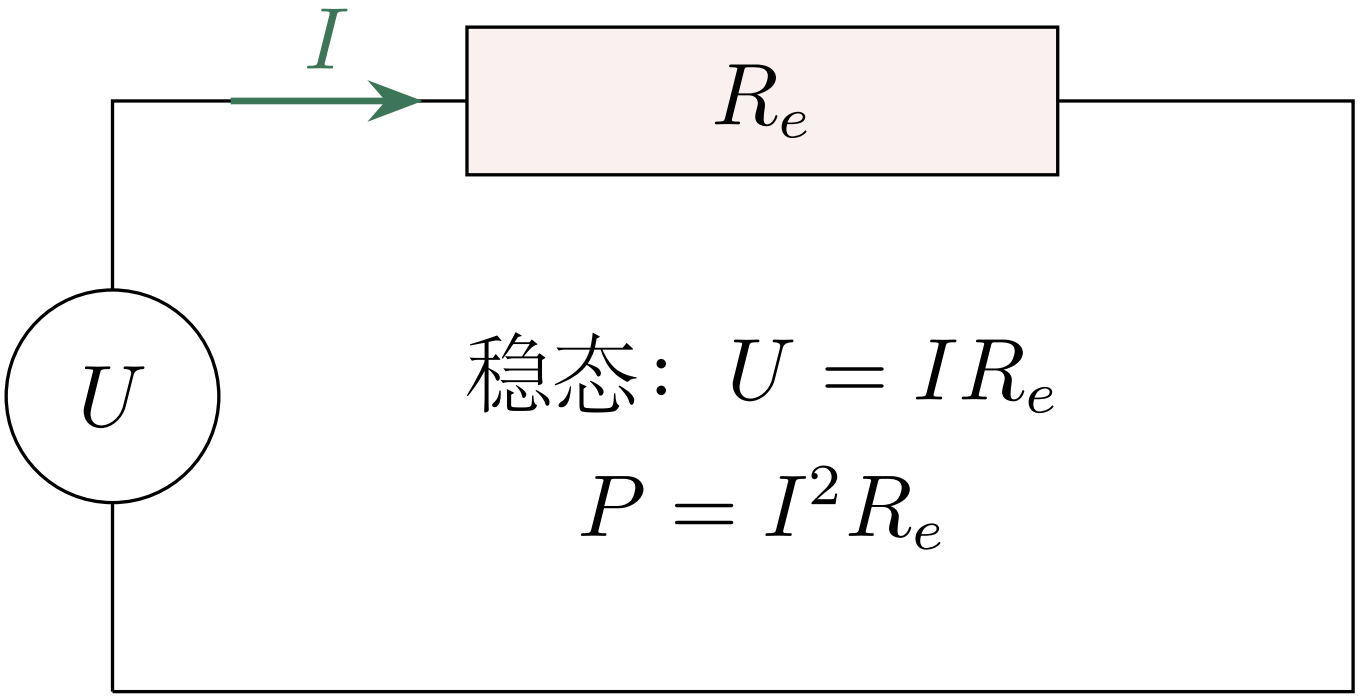


同一铁块，两处向上吸力相加



铜线绕截面，磁通沿铁芯

背侧虚线、线径及匝距均为示意



磁路激励由同一绕组的 NI 决定